

Igor Uchoa

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<http://igoruchoa.github.io>

PhD candidate at the Atmospheric and Oceanic Science Department of the University of Maryland, College Park (UMD) working on air-sea interaction at submesoscale using observations and coupled model simulations. Experienced in data analysis of oceanographic observational data and numerical simulations.

Education

Academic Qualifications.....

- **PhD Student Atmospheric and Oceanic Science** **2021 - Present**
University of Maryland, College Park, Maryland
- **M.Sc. Physical Oceanography** **2017 - 2019**
Oceanographic Institute of University of São Paulo, São Paulo, Brazil
- **B.S. Oceanography** **2012 - 2016**
Federal University of Ceará, Fortaleza, Brazil

Research

- **Graduate Assistant** **College Park, MD**
University of Maryland, College Park
Advisor: Jacob Wenegrat
Aug 2021 - Present
- **Oceanographic Cruise Coordinator Assistant** **São Paulo, Brazil**
University of São Paulo
Coordinator: Prof. Ilson C. A. Silveira
7-21, Dec 2019
- **Petróleo Brasil (PETROBRAS) / University of São Paulo** **São Paulo, Brazil**
Position: Oceanographer, Coordinator: Prof. Ilson C. A. Silveira
Jul 2019 – Jul 2020
Main Activities: Oceanographic and Meteorological data processing; Model outputs and reanalysis processing; Plotting and sketch illustration for Data Science; Scientific writing for reports, papers, and presentations.

Teaching

- **Instructor** **University of Maryland College Park**
Weather and Climate Laboratory
Aug 2025 - May 2026
- **Teacher Assistant** **University of Maryland College Park**
Weather and Climate
Instructor: Jeffrey Henrikson
Aug 2025 - May 2026
- **Instructor** **Neuromatch Academy - Remote**
Climatematch
Coordinator: Nick Halper
Jul 2024
- **Teaching Assistant** **University of São Paulo**
Course: Dynamic Oceanography II
Coordinator: Prof. Ilson C. A. Silveira
Jul 2018 – Dez 2018

Publications

Papers.....

- I. **Uchoa**, J. O. Wenegrat, A. Kinsella, I. M. Leyba, L. W. O'Neill, and L. Lenain. Observed rapid adjustment of the atmospheric boundary layer to submesoscale sea surface temperature fronts. *Proceedings of the National Academy of Sciences*, 123(30):e2605662123, 2026
- L. Renault, C. Conejero, J. Wenegrat, and I. **Uchoa**. Interplay of submesoscale current and thermal feedbacks: A seasonal perspective in the gulf stream. *Geophysical Research Letters*, 52(21), 2025
- I. **Uchoa**, J. Wenegrat, and L. Renault. Sink of eddy energy by submesoscale sea surface temperature variability in a coupled regional model. *Journal of Physical Oceanography*, 55, 2025
- J. T. Farrar and **et al.** S-MODE: The sub-mesoscale ocean dynamics experiment. *Bulletin of the American Meteorological Society*, 2025
- L. Renault, M. Contreras, P. Marchesiello, C. Conejero, I. **Uchoa**, and J. Wenegrat. Unraveling the impacts of submesoscale thermal and current feedbacks on the low-level winds and oceanic submesoscale currents. *Journal of Physical Oceanography*, 54(12):2463–2486, 2024
- L. Nuijens and **et al.** The Air-Sea Interaction (ASI) submesoscale: Physics and impact. In *Lorentz Workshop*, 2024
- F. Santos, L. Araújo, I. **Uchoa**, R. Lourenço, S. Taniguchi, C. Martins, R. Nagai, I. Wainer, M. Mahiques, R. Figueira, and M. Bícigo. High-resolution UK'37 sea surface temperature reconstruction in the southwestern Atlantic Ocean during the Holocene. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 654:112452, 2024
- I. **Uchoa**, I. T. Simoes-Sousa, and I. C. da Silveira. The Brazil Current mesoscale eddies: Altimetry-based characterization and tracking. *Deep Sea Research Part I: Oceanographic Research Papers*, 192:103947, 2023
- A. C. T. Bonecker, M. Katsuragawa, M. S. de Castro, C. Namiki, M. de Lourdes Zani-Teixeira, I. **Uchoa Farias**, and I. C. A. d. Silveira. Seasonal variability of ichthyoneuston assemblage on the continental shelf and slope of the Southwest Atlantic Ocean, Brazil (20–23 S). *Journal of applied ichthyology*, 35(3):655–671, 2019

Book Chapter.....

- I. C. A. da Silveira, D. C. Napolitano, and **Farias Uchoa, I.** *Water Masses and Oceanic Circulation of the Brazilian Continental Margin and Adjacent Abyssal Plain*, pages 7–36. Springer International Publishing, Cham, 2020

Scientific Cruises & Field Programs

- **S-MODE - Intensive Operating Period 1**
R/V *Bold Horizon*, P.I.: Tom Farrar

US
10-30 Oct 2022

- **Projeto VT-Dyn**
R/V Alpha Crucis, P.I.: Ilson Silveira

Brazil
7-21 Dec 2019

Advising

- **Co-advisor**
Megan Brown - Undergraduate Research Project 2024

Guest Lecture

- Introduction to Oceanography - Weather and Climate (AOSC 200) - April 2026

Invited Talks

- **Observed air-sea thermal feedback at submesoscale fronts**
American Geophysical Union Annual Meeting, Washington DC 2024
- **Observed rapid adjustment of the atmospheric boundary layer to submesoscale SST fronts**
OASIS Air-Sea Webinar, Virtual 2026

Activities, Outreach and Service

- **2023-Present:** Reviewer for Journal of Physical Oceanography, Journal of Geophysical Research - Oceans, Journal of Southern Hemisphere Earth Systems Science
- **2024-2025** Letters for a Pre-scientist (LPS) - Volunteer
- **2024-Present** Communications for Science Policy Group - University of Maryland, College Park
- **2023-2024** Student Seminar - Atmospheric and Oceanic Science Department, University of Maryland - College Park

Additional Training

- **2022-2023** - Life Skills for Young Scientists, Planetary Science Institute

Achievements

- **2025** - Richard Payne Graduate Award - University of Maryland, College Park
- **2024** - Green Fund Scholarship (Green Fund Endowed Graduate Fellowships in Earth Sciences and Climate Change) - University of Maryland, College Park
- **2024** - Group Achievement Award Submesoscale Ocean Dynamics Experiment (S-MODE) - NASA
- **2021** - College of Computer, Mathematical, and Natural Sciences Deans Fellowship - University of Maryland, College Park
- **2013** - Science Without Borders Scholarship - Brazil Government